

Bonus Assignment_1

Overview:

In this assignment you will create estimates for the different tasks needed for the project plan for the software product requested in the Requirements Specification document for **Chicago WideCast Smart-Home Services**.

Instructions:

1. Create venv that has Python 3.10
2. Execute every experiment requested for every model on every platform using the frameworks listed in the provided table below:

Experiment Number	Platform	Model		Framework
1	Replicate [Cloud]	meta/meta-llama-3-70b-instruct	HuggingFaceEmbedding BAAI/bge-small-en-v1.6	LangChain/LangGraph
2	OpenAI [Cloud]	gpt-4o-mini	text-embedding-3-small	LangChain/LangGraph
3	Replicate [Cloud]	deepseek-ai/deepseek-r1	HuggingFaceEmbedding BAAI/bge-small-en-v1.6	LangChain/LangGraph

Requirements:

1. Use the provided PDF document for the requirements specification document for **Chicago WideCast Smart-Home Services**.
2. Instruct the GenAI model/platform to create unique TAGS/NUMBERS for the requirements/use-cases of Widecast. See **Appendix A** for a sample output.
3. Instruct the GenAI model/platform to use **SampleProjectTasksEstimates.pdf** when creating the tasks and estimates for the request project plan.
4. Instruct the GenAI model/platform to consider the following roles when creating the tasks and estimates: Project Manager, Requirements Engineer, System Engineer, Software Engineer, Test Engineer, and Documentation Engineers.
5. Create an AI Agent to represent every role that will generate the tasks and the estimates: Project Manager, Requirements Engineer, System Engineer, Software Engineer, Test Engineer, and Documentation Engineers. Every Ai Agent for every role will generate the tasks and the estimates.
6. Do NOT hard code tasks or estimates in your sources code. You will get ZERO for that. Rather, your AI Agents will feed the PDF document to the GenAI model and communicate/synchronize with the project manager and other AI Agents when generating the tasks/estimates.

Appendix A:

TAGED/NUMBERED Requirements generated by OpenAI/gtp-4o-mini

Requirements & Use Cases

1. REQ-001: Purchase TV Plan

- *Use Case:* A customer selects a TV plan (Basic, BasicPlus, or Ultimate) and subscribes based on the available contract options.

2. REQ-002: Cancel Internet Subscription

- *Use Case:* A customer with an active data plan (WiFi SpeedLane or LightLane) requests cancellation following contract rules.

3. REQ-003: Upgrade/Downgrade Subscription

- *Use Case:* A customer modifies an existing TV, Data, or Streaming plan to a higher or lower tier.

4. REQ-004: Order PPV Event

- *Use Case:* A customer purchases a pay-per-view event (live sports or movies) based on their TV plan eligibility.

5. REQ-005: Register New Customer Account

- *Use Case:* An Account Specialist creates a new customer account, including billing and service preferences.

6. REQ-006: Pay Monthly Bill

- *Use Case:* A customer securely processes a payment for their subscribed services.

7. REQ-007: Create & Assign Support Ticket

- *Use Case:* An Account Specialist logs a support request and assigns it to a Technician for resolution.

8. REQ-008: Change Payment Method

- *Use Case:* A customer updates their credit card or billing details for future payments.

9. REQ-009: Cancel TV Subscription

- *Use Case:* A customer requests cancellation of their TV plan while following contract obligations.

10. REQ-010: Activate Home Security System

- *Use Case:* A customer enables security cameras and remote door locking features.

11. REQ-011: Modify Customer Account Information

- *Use Case:* A customer updates personal details like email, phone number, or address.

12. REQ-012: Technician Closes Support Ticket

- *Use Case:* A Technician marks a reported issue as resolved after troubleshooting.

13. REQ-013: Automate AI Chat Support for Service Inquiries

- *Use Case:* The AI assistant answers customer queries about available plans, billing, and troubleshooting.

14. REQ-014: Schedule a Service Appointment

- *Use Case:* A customer books an installation or repair visit with a Technician.

15. REQ-015: Request a Refund for a PPV Event